

MOHD HAMMAD KHAN

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PROFESSIONAL SUMMARY

Result-driven B.Tech Computer Science & Engineering student (Graduation: June 2026) with specialized proficiency in Python development, network security, and backend systems. Proven ability to architect scalable telecom solutions, optimize performance by 15-20%, and implement real-time security monitoring systems. Experienced IT trainee skilled in REST API design, database optimization, and asynchronous programming. Strong foundation in SDLC, Agile methodologies, and practical system implementation. Seeking roles as Backend Engineer, Python Developer, or Software Engineer.

TECHNICAL SKILLS

Programming Languages: Python | C++ | SQL | Bash Scripting

Backend & Web Development: REST API Design | Socket Programming | TCP/IP Protocols | Asynchronous Programming | Network Architecture | Flask | FastAPI

Frameworks & Libraries: Pygame | Scapy | NumPy

Cybersecurity & Network Analysis: Intrusion Detection Systems (IDS) | Network Threat Detection | Packet Analysis | Real-time Monitoring | IoT Security Fundamentals | TCP/IP Protocol Implementation

Databases & Performance Optimization: MySQL | SQL Query Optimization | Database Indexing | Connection Pooling | Large-scale Data Processing

Software Engineering Practices: SDLC | Agile Development | Scrum Methodology | Code Optimization | Performance Profiling (cProfile) | Git/GitHub Version Control | Testing Frameworks | Advanced Testing & CI/CD Integration (pytest, GitHub Actions)

Core Computer Science: Data Structures & Algorithms | Object-Oriented Programming (OOP) | Operating Systems | Computer Networks | System Architecture Basics

Development Tools & Environments: Linux (Ubuntu) | Git | GitHub | VS Code | Command-line Tools | Memory Profilers | SSH/TLS Protocols | Cloud Deployment & Container Technologies (Docker)

PROFESSIONAL EXPERIENCE

Summer IT Trainee

Oranix Technologies Pvt. Ltd.

June 2025 - July 2025

2-month Internship

Engineered production-level Python applications for high-availability telecom solutions serving 1,000+ concurrent connections, utilizing REST API design and asynchronous programming patterns.

- Performance Optimization & System Design:** Increased application execution performance by 15% through advanced code profiling using cProfile and algorithmic optimization.
- Reduced memory consumption by 20% via memory-efficient data structures and connection pooling implementation.
- Deployed MySQL query optimization with indexing strategy, reducing response times from 500ms to 250ms.
- Real-time Data Management:** Managed and analyzed real-time network data feeds during comprehensive software testing cycles.
- Collaborated with 3-person cross-functional team using Agile/Scrum methodology (2-week sprints) and Git version control.
- Conducted technical code reviews on GitHub, ensuring code quality and architectural best practices.
- Technical Stack: Python | MySQL | Linux | REST API Principles | Git | Agile Methodology*

PROJECTS

Intrusion Detection System (IDS) for IoT Devices

June 2025 | 4 weeks

GitHub: github.com/hammad12011/IDS-IoT-Security

Designed and deployed a real-time Intrusion Detection System using Scapy to monitor network traffic and detect suspicious activities in IoT environments.

- Processed and monitored 1,000+ concurrent network connections simultaneously with sub-second response times.
- Securely indexed and parsed 100,000+ security logs into MySQL database with optimized indexing, achieving <100ms data retrieval latency.
- Implemented automated alerting mechanisms for real-time threat notifications.
- Executed database optimization including query optimization, connection pooling, and data normalization.
- *Technologies: Python | Scapy | MySQL | Linux | TCP/IP Protocols | Socket Programming | Real-time Monitoring*

Python Interactive Gaming Application

May 2025 | 3 weeks

Developed an interactive command-line gaming application using Object-Oriented Programming principles with modular class architecture and state management.

- Integrated Pygame library for graphics rendering and user input handling with responsive mechanics.
- Achieved 25% runtime performance improvement through algorithmic optimization and collision detection refinement.
- Optimized memory usage via efficient data structure selection and garbage collection awareness.
- Implemented unit testing framework for game logic validation and edge case handling.
- *Technologies: Python 3.10+ | Pygame | OOP Design Patterns | Algorithm Optimization | Testing*

ACHIEVEMENTS & CERTIFICATIONS

- **IBM Certification:** Python Fundamentals for Data Science Professionals | 2025
- **IBM CEPYT Certificate:** Certified Entry-level Python & Technology Fundamentals | 2024
- **National Recognition:** Merit Certificate - Viksit Bharat Young Leaders Dialogue (VBYLD) 2026 (Selected among 500+ candidates nationally)
- **Technical Portfolio:** Multiple academic and standalone software engineering projects with full GitHub repositories and working implementations

EDUCATION

Bachelor of Technology (B.Tech) in Computer Science & Engineering

Graduation: June 2026

Advanced Coursework SGPA: 8.70/10 (Final Semester)

Integral University, Lucknow | Academic Standing: First Division